
AWS Compute Optimizer – Overview

What is AWS Compute Optimizer?

AWS Compute Optimizer is a **free AWS service** that uses **machine learning** to analyze your **AWS resource usage** and recommend the **most optimal instance types** and configurations to improve **performance** and **reduce cost**.

 It helps you answer:

“Am I using the right instance size and type for my workload?”

What It Optimizes

AWS Compute Optimizer supports the following resources:

Resource Type	Optimization Area
EC2 Instances	Instance size/type, CPU/RAM balance
EBS Volumes	Volume type and performance
Lambda Functions	Memory size, duration
Auto Scaling Groups	Instance type for groups

How It Works

1. **Monitors** resource utilization using **CloudWatch metrics**
2. **Analyzes** historical data (up to 14 days)
3. **Recommends**:

- **Right-sized** EC2 instances
- **Volume changes** for EBS
- **Lambda memory tuning**

4. Recommendations include:

- Performance risk (low/medium/high)
- Cost impact
- Migration effort

Example: EC2 Recommendation

Metric	Current Instance	Recommendation
CPU Utilization	t2.medium (10%)	t3.small (cheaper)
Memory Usage	Low	Smaller instance size
Performance Risk	Low	Safe to downsize

Accessing Compute Optimizer

- Console: [AWS Compute Optimizer Console](#)

CLI:

aws compute-optimizer get-ec2-instance-recommendations

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Key Features

Feature	Description
ML-powered insights	Learns from real usage patterns
Cross-service support	EC2, EBS, Lambda, Auto Scaling
Cost savings estimate	Shows potential cost impact
Performance risk scores	Highlights reliability trade-offs
Resource filters	Filter by region, instance type, risk level

Cost

Free

You only pay for the AWS services you're running — Compute Optimizer itself is free of charge.

Best Practices

- Regularly review Compute Optimizer reports
 - Use recommendations to **resize EC2 instances**
 - Combine with **Savings Plans or RI** purchases for deeper savings
 - Enable **enhanced infrastructure metrics** for more accurate analysis (paid option)
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Related Tools

Tool	Purpose
Trusted Advisor	Broader checks (security, fault tolerance)
Cost Explorer	Visualize usage and cost trends
Compute Optimizer	Deep ML-based optimization for compute

Summary

Aspect	Details
What	ML-powered recommendations for compute resources
Why	Save costs, improve performance
Supports	EC2, EBS, Lambda, Auto Scaling
Cost	Free
Access	Console, CLI, SDK
